

5.3

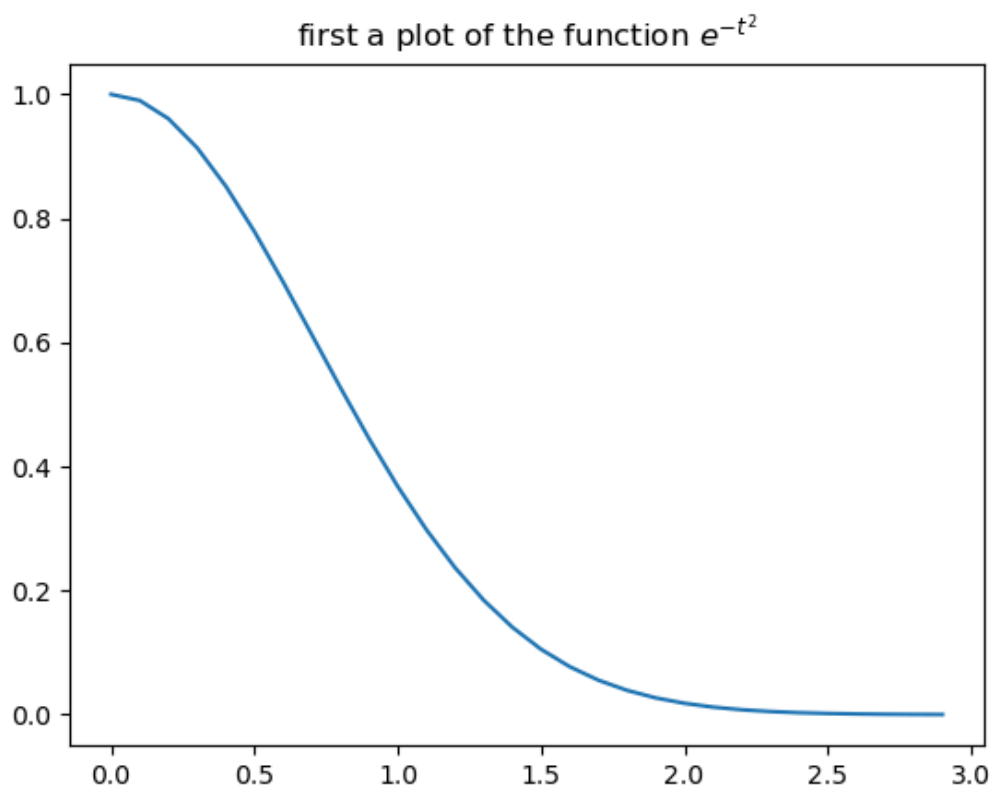
```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

```
In [3]: def g(t):
        return(np.exp(-(t**2)))
t = np.arange(0,3,0.1)

fig, ax = plt.subplots()
ax.plot(t, g(t))
ax.set_title(r'first a plot of the function  $e^{-t^2}$ ')
```

Out[3]: Text(0.5, 1.0, 'first a plot of the function e^{-t^2} ')

Figure



```
In [4]: def simpl(func, a, b, steps):
        if steps%2!=0:
            steps = steps + 1
            print('i added one step')
        dx = (b-a)/steps
        x = np.linspace(a, b, steps+1)
        y = func(x)
        s = dx/3*(y[0]+y[-1])
```

```

s = s + dx/3*(4*np.sum(y[1:steps:2]) + 2*np.sum(y[2:steps-1:2]))
return(s)

def f(x):
    def g(t):
        return(np.exp(-(t**2)))
    integral = 0.0
    integral = simpl(g, 0 , x, 1000)
    return(integral)

x = np.arange(0,3,0.1)
y = []
for i in x:
    y.append(f(i))

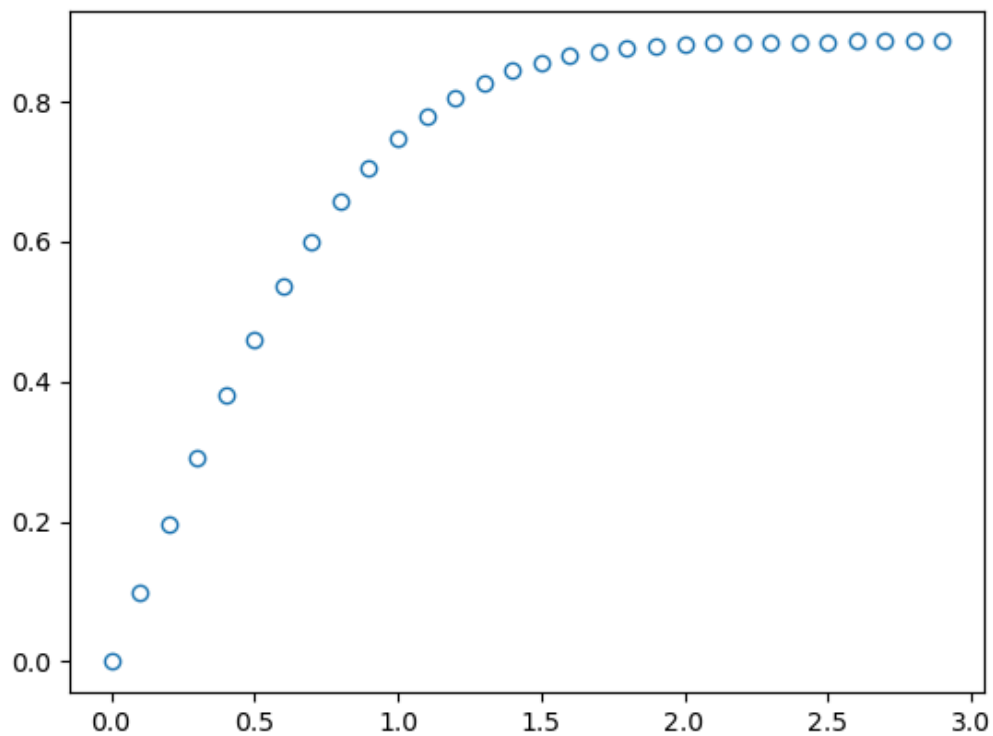
fig, ax = plt.subplots()

ax.plot(x, y, 'o', mfc='none')

```

Out[4]: [<matplotlib.lines.Line2D at 0x7246127962d0>]

Figure



In []:

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In [5]: `x = np.arange(0,3.1,0.1)`

In [6]: `x`

Out[6]: `array([0. , 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1. , 1.1, 1.2,
1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2. , 2.1, 2.2, 2.3, 2.4, 2.5,
2.6, 2.7, 2.8, 2.9, 3.])`

In []: